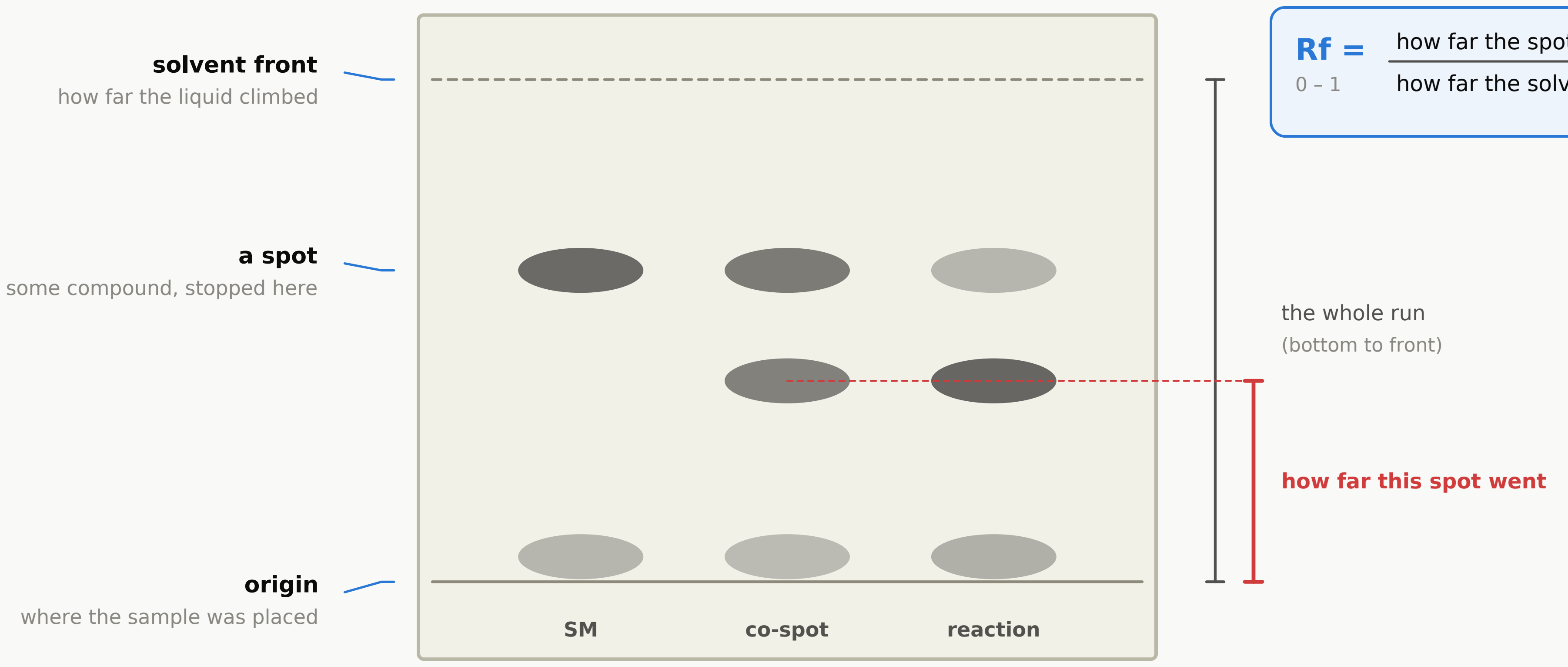
A chemist reads one of these in five seconds

A sheet of silica. A line where the sample started, a line where the solvent stopped, and some dark patches in between.

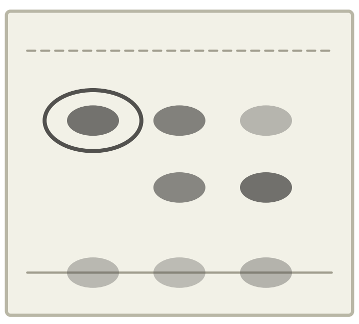


$$Rf = \frac{\text{how far the spot went}}{\text{how far the solvent went}}$$

0 – 1

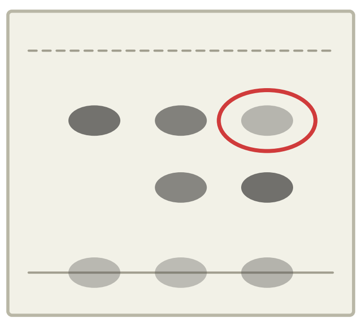
AND THIS IS WHAT THEY DO WITH IT

1 look at the reference lane



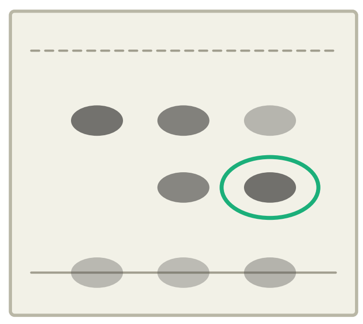
this is what I started with

2 has it gone?



faint at the same height — nearly consumed

3 is there something new?



strong, and matched in the co-spot lane

4 verdict

- starting material: nearly gone
- product: formed, now major
- nothing unexplained

give it one more hour

elapsed time: about five seconds

THE BAR

That is the incumbent. Free, instant, and very good at the part that matters. Anything we build is competing with it.

It has already been built. About thirty times.

Starting in 2007, with a high-school science-fair project that got published in a teaching journal.

2007
first published photo-to-chromatogram tool

30+
distinct tools that read a plate from an image

~1,000
installs of the best of them, worldwide

0
that say what any spot actually is

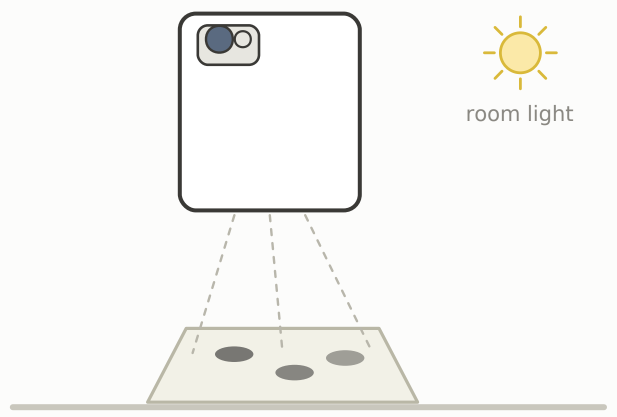
READ THE EMPTY CORNER

Everything cheap sits along the bottom: it finds the spots and stops. Everything that climbs higher had to buy hardware. The top-left corner is empty.

Six ways to get light off a plate

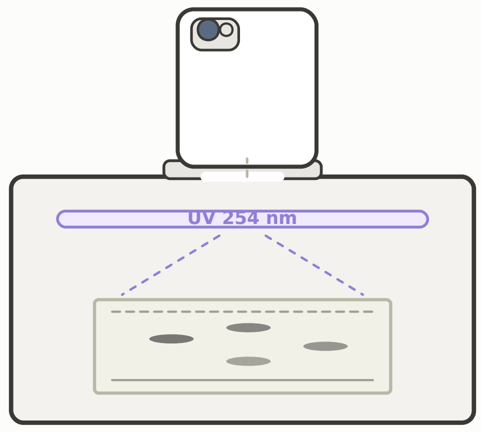
Each one removes a bit more of the mess — and costs a step more. The software that follows is almost identical in all six.

A phone, on the bench
free, and unrepeatable



nothing is controlled

A phone and a box
$36–78 in parts



a shoebox, or a 3D-printed one

A flatbed scanner
one you already own

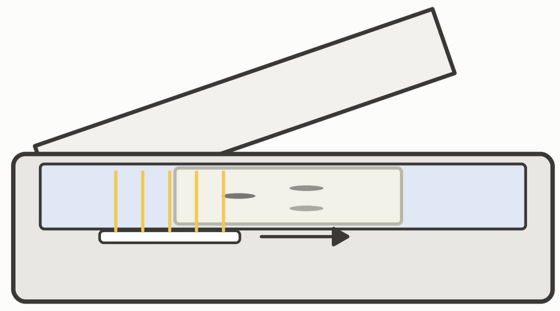
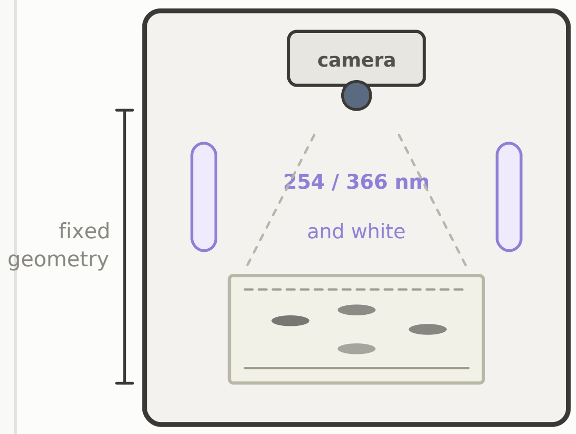


plate laid face-down on the glass

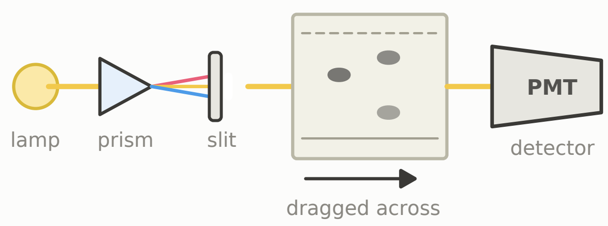
the scanner's own lamp is the light source

An imaging cabinet
capital equipment



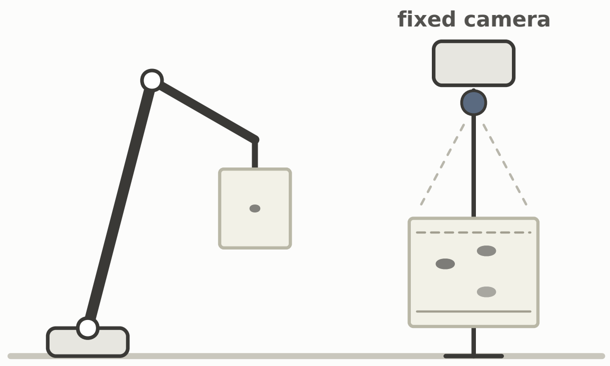
the box removes the variability

A slit densitometer
no camera at all



no camera, no photograph — a physical measurement

A robot and a camera
removes the human



a robot spots and moves every plate the same way

4,944 measurements published, no images

AND WHAT THE SOFTWARE DOES NEXT, IN EVERY SINGLE CASE

1. **find the lanes** — almost always by hand
2. **turn each lane into a curve** — column by column
3. **pick a cut-off height** — a number the user guesses
4. **report position and darkness** — never a name

THE SHAPE NEVER CHANGED

Eighteen years, six kinds of hardware, three continents — and the answer that comes out is always a number attached to a position.

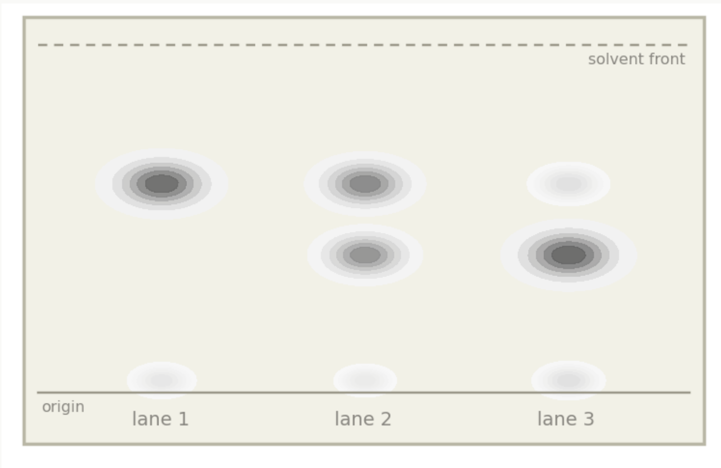
A spot has no name

It has a position and a darkness. Both are in the file. Nothing else is.

THE SAME PIXELS, THREE DIFFERENT ANSWERS

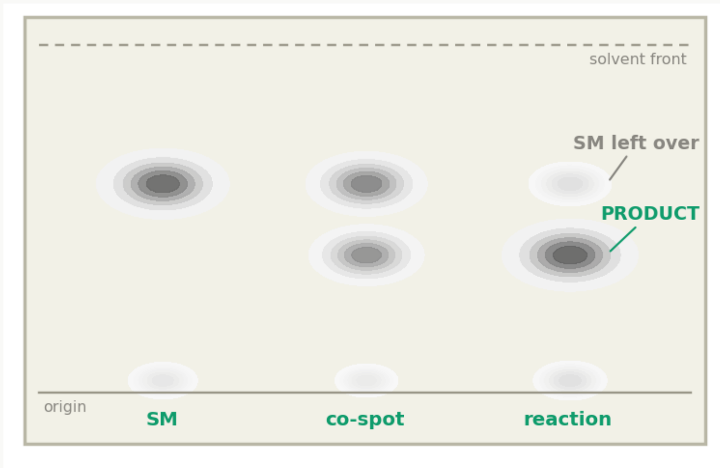
A spot carries a position and a darkness. It does not carry a name. Naming it needs the chemist's context — which is why measuring and interpreting are two separate jobs.

WHAT THE CAMERA RECORDS



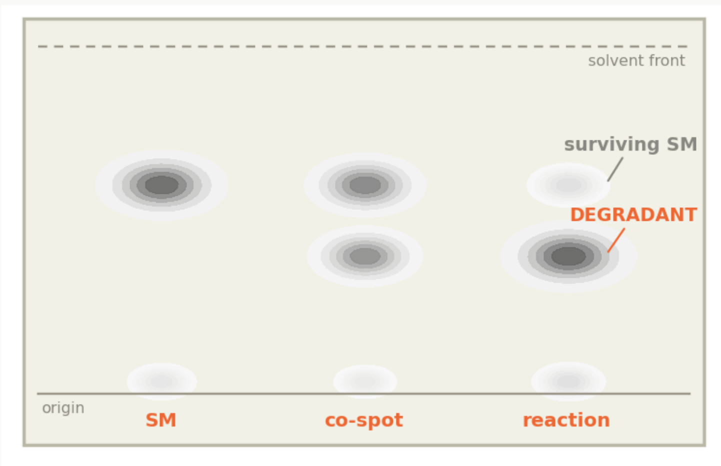
Three lanes. Eight dark regions. Each has a position and a darkness, and that is all the file contains.

READING 1 — it worked



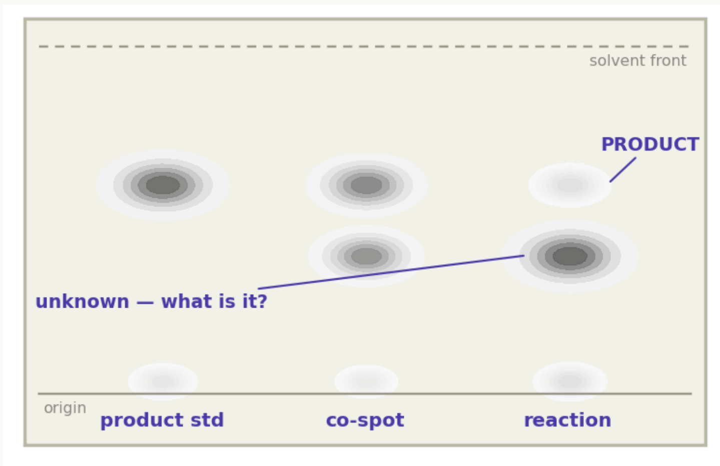
The new spot at 0.41 is the product; the faded 0.62 is unreacted starting material. Nearly done — one more hour.

READING 2 — it went wrong



Identical picture. If that new spot is a break-down product, the reaction failed and the material is being eaten. Stop.

READING 3 — lanes swapped



If lane 1 held the product standard, the 0.62 spot is the product and the big new spot is something nobody asked for.

Which lane held what

Not in the picture. The chemist knows; the file does not.

What reaction this is

Not in the picture either — and it decides every reading above.

What was expected to form

Without it, a new spot is just a new spot.

THE REAL REASON

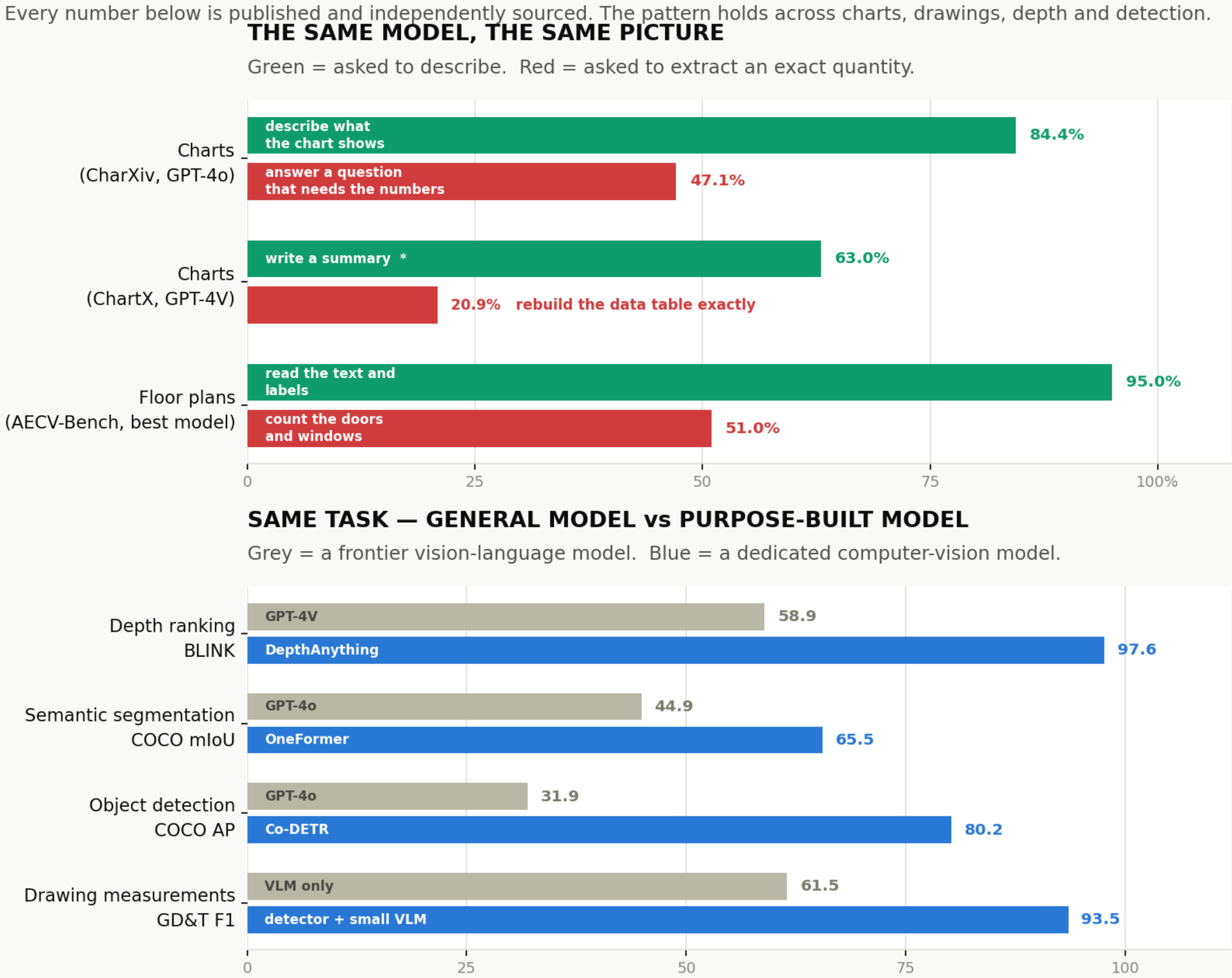
So the tools did not stop from lack of ambition. They stopped because the next step is not an image problem at all.

Half of it, very well

Vision models describe brilliantly and measure badly. That is measured, not assumed.

VISION MODELS DESCRIBE WELL AND MEASURE BADLY

Every number below is published and independently sourced. The pattern holds across charts, drawings, depth and detection.



* ChartX rates description quality 0-5; GPT-4V scored 3.17, shown here rescaled to 63%. Its data-table score is a direct percentage.

Sources: CharXiv (NeurIPS 2024) · ChartX arXiv:2402.12185 · AECV-Bench arXiv:2601.04819 · BLINK (ECCV 2024) · GPT-4o vision evaluation arXiv:2507.01955 · Khan et al. RCIM 2025.
Separately, MeasureBench (arXiv:2510.26865) asked frontier models to read gauges and dials: the best scored 30.2% at publication in October 2025, 41.7% on the live leaderboard today.

AND EXACTLY THE SAME SPLIT ON OUR OWN PLATES

	classical software	a language model
Read the handwritten lane labels	0 of 14	all correct
Find the spots and measure them	6 of 6	misses faint ones, drifts
Say which spot is the product	cannot	correct, given the context

THE USEFUL PART

The two columns are near-opposites. Whatever gets built, the measuring and the naming should not be done by the same thing.

Solved, empty, and moving

Three things are true at once, and they pull in different directions.

SOLVED

Measuring a plate

- Thirty-plus tools already do it, going back to 2007.
- Several are free; one drives a purification system.
- A phone gets within 2–5% of a densitometer in a lab.
- It is 15–30× less sensitive, which rarely matters for a reaction.
- And almost nobody uses any of them.

EMPTY

Naming what is on it

- No product reads a plate in the context of its reaction.
- The most advanced thing on the market classifies whole samples, not spots.
- Nothing published measures a real bench plate and names what is on it.
- There is no public dataset of labelled plate images anywhere.
- And no object-detection paper for TLC spots exists at all.

MOVING

Who the buyer is

- A chemist does it in five seconds and never needed help.
- A robot cannot do it at all.
- About $4bn went into robots that run reactions in eighteen months.
- None of them uses TLC as a sensor — they use NMR, HPLC, MS.
- Which is either the gap, or the reason there is no gap.

AND THE HONEST COUNTERWEIGHT

The one large field trial of automated plate reading raised detection of bad medicines from 35% to 62.5% — and dropped specificity to 75.2%, flagging one good medicine in four. Its own authors concluded it was not fit to deploy. Anyone who says this problem is easy has to explain that result first.

THE WHOLE THING IN ONE SENTENCE

Measuring a TLC plate is finished work that people give away for free. Naming what is on it, in the context of the reaction it came from, is what nobody sells.

THREE THINGS WORTH WATCHING

Vision models

A benchmark asking them to read gauges went from 30% to 42% best score in under a year.

CAMAG's AI add-on

Released February 2026 — the first commercial product to sell anything above measurement.

Autonomous labs

If they settle on benchtop NMR for reaction feedback, the machine buyer never appears.